Sumanth Reddy Cherupally

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Work Experience

Inscope | Founding AI Engineer | Remote **Oct 2025 – Present**

- Designed and deployed high-precision agentic AI systems for financial data processing, including multi-step reasoning, tool use, workflow orchestration, and reliability-focused evaluation.
- Built and owned 0→1 AI infrastructure across Node.js and FastAPI backends, model integration, and production-grade APIs, delivering scalable, low-latency AI systems under ambiguity.
- Drove end-to-end product development with the founding team, owning mission-critical AI features, full-stack delivery (React), and core platform intelligence.

Barclays | Software Development Engineer | Whippany, NJ **Aug 2023 – Oct 2025**

- Engineered Control-Check, a robust data framework for financial reporting systems processing 20+ million rows, implementing algorithm optimization techniques and advanced statistical methods that reduced downstream data issues by 50% with no additional processing time while providing scalable validation and quality assurance for large datasets.
- Built and maintained scalable web applications using Django and Streamlit frameworks, ensuring system performance and reliability for enterprise-scale deployment and data visualization needs.
- Developed data-intensive Python applications with automated deployment infrastructure using Chef configuration management and GitLab CI/CD pipelines, streamlining release processes and eliminating manual deployment errors across multiple production environments.

TuSimple, Inc. | Software Development Intern - **Onboard Runtime Team** | San Diego, CA **Jun 2022 – Dec 2022**

- Developed a GPU Tracing Tool using CUDA and performance analysis algorithms to identify and resolve runtime anomalies in autonomous vehicle systems, reducing developer debugging time by 75%.
- Optimized database schema and query algorithms, achieving 40% storage reduction and 50% query time improvement for large-scale vehicle data retrieval systems.

Research

Existential Robotics Lab, UCSD | Prof. Nikolay Atanasov, ECE Jan 2022 – Aug 2023

- Implemented deep reinforcement learning models with TensorFlow for end-to-end autonomous vehicle control, using policy gradient algorithms to optimize performance in complex vision-based environments.
- Led development of a multi-agent SLAM system employing distributed sparse Gaussian processes for real-time 3D environment mapping, optimizing for computational efficiency and accuracy.
- Demonstrated expertise in Reinforcement Learning, Robotic perception, computer vision, and C++ programming.

Lab for Video and Image Analysis, IIT Hyderabad | Prof. Sumohana S. Channappayya, EE Aug 2020 – May 2021

- Designed and trained advanced neural network architectures (ResNet variants) for high-precision object recognition, achieving 95% recall (from 76%) at constant precision for occlusion detection.
- Developed algorithms for monocular depth estimation enabling efficient processing of complex 3D datasets with applications in scene understanding.
- Optimized the model pipeline to facilitate real-world deployment, and provided a mechanism for human intervention and tuning of model hyper-parameters to further improve performance.

Practical Networking and Blockchain Lab, IIT Hyderabad | Prof. Kotaro Kataoka, CSE May 2019 – Apr 2021

- Introduced a novel blockchain protocol optimized for IoT devices, focusing on scalability and resource efficiency. Developed communication protocols using **Golang** with multi-threading and channels for concurrency control. Published two papers in the Journal of Network and System Management and ICOIN 2021. Find details in the Publications section.
- Open-sourced the final system.
  Repositories: github.com/sumanthcherupally/LSDI_GW, github.com/sumanthcherupally/LSDI_SN
- Engineered a distributed 200-node P2P network testbed leveraging multiple servers and Docker containers for extensive blockchain network validation and performance testing. Demonstrated the system at CEATEC 2019 exhibition in Japan as IIT Hyderabad's representative, running live blockchain operations on Raspberry Pi infrastructure.

Education

University of California, San Diego, US | MS in ECE (Robotics and Deep Learning) **Sept 2021 - July 2023**
Indian Institute of Technology Hyderabad - IIT Hyderabad | B.Tech in Electrical Engineering **Jul 2017 - June 2021**
Coursework: Deep learning, Reinforcement Learning, Advanced Computer Vision, Linear Algebra, Data structures and algorithms.

Skills

Machine Learning: LLMs, Deep Learning, Reinforcement Learning, Computer Vision, Neural Networks, TensorFlow, PyTorch
Programming: Python, C++, Golang, CUDA.
Frameworks/Tools: ROS, Docker, SQL, Git, Pybullet, MATLAB, Debugging/Performance Analysis.

Projects

- **Transformer Architecture Comparison Framework** - Developed a PyTorch-based framework for comparing different transformer architectures used in modern LLMs, implementing multiple attention mechanisms and positional encodings. Created a modular system for evaluating performance trade-offs across different LLM architectures using WikiText-2 dataset. Implemented GenAI solutions through optimized transformer architectures, contributed to ML infrastructure by building reusable components, and enhanced performance through custom attention mechanisms and efficient data processing pipelines.
- **Parallel Computation** - Achieved 1100x performance improvement for matrix multiplication on NVIDIA GPUs using CUDA programming. Implemented cache optimization techniques and instruction-level parallelism for 7x CPU performance gain. Designed algorithms for efficient communication between multiple cores using MPI for distributed computing applications.
- **MCP Server Development** - Developed a Python-based MCP (Model Context Protocol) server that integrates Gmail, Google Calendar, and Google Maps APIs to enable AI assistants to manage emails, schedule events, and provide location services through natural language commands. Built 7 specialized tools using FastAPI and OAuth 2.0 authentication, demonstrating expertise in API integration, microservices architecture, and modern software development practices.
- **Deep Learning models** - Implemented various Deep CNN architectures and LSTMs for Image segmentation and Image captioning tasks. Eg: ResnNet, UNet, PSPNet etc.
- **Neural Radiance Fields (NeRF) for View Synthesis** - Implemented and optimized the canonical NeRF architecture and variants (Mip-NeRF, Pixel-NeRF) for novel view synthesis. Developed efficient algorithms for 3D scene representation learning from 2D images, optimizing neural network training for high-quality rendering.

Publications

- **S. R. Cherupally**, S. Boga, P. Podili and K. Kataoka, "Lightweight and Scalable DAG based distributed ledger for verifying IoT data integrity" 2021 International Conference on Information Networking (ICOIN), 2021. [link]
- Prashanth Podili, **S. R. Cherupally**, Srinivas Boga, Kotaro Kataoka, "Inter-domain prefix and route validation using fast and scalable DAG based distributed ledger for secure BGP routing" Journal of Network and Systems Management, Special Issue on Blockchains and Distributed Ledgers in Network and Service Management. [link]